DIMENSION OF A GROUPOID OVER A FIELD IS CONSTANT AND TRANSDEGREES VIA S AND T AGREE

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Editorial note. Part (3) is asserted as 'a formal consequence of the results obtained in the discussion above' and is carried as an explicit obligation. The move-point dependence is declared as external.

Allowed standard prerequisites (editorial).

- Dimension formula at a point. Morphisms, lemma dimension-fibre-at-a-point (cross-chapter, named). Etale-local computation dim_r(R) = dim(O_{R,r-bar}) + trdeg.
- Dimension invariance under field base change. Morphisms of Spaces, lemma dimension-fibre-after-base-change (cross-chapter, named). Dimension at a point unchanged under the two field extensions.
- Point explanation and moving. lemma-groupoid-on-field-explain-points (structural use) and lemma-groupoid-on-field-move-point (same chapter, NOT carded). Reduces (1) to the identity point.

Source reference. Original lemma 04LR: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 977–977. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04LS: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1015–1015. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04LQ: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 937–937. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02FX: focused-seeds/geometry/S9-smg-groupoid-on-field-locally-finite-type-dimension/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 06DY: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 859–859. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NS: focused-seeds/geometry/S9-smg-groupoid-on-field-locally-finite-type-dimension/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 06DY assume s, t are locally of finite type. For all $r \in |R|$*

1. $\dim(R) = \dim_r(R)$,
2. *the transcendence degree of r over* $\mathrm{Spec}(k)$ *via s equals the transcendence degree of r over* $\mathrm{Spec}(k)$ *via t, and*
3. *if the transcendence degree mentioned in (2) is* 0, *then* $\dim(R) = \dim(\mathcal{O}_{R,\overline{r}})$.

Proof. Let $r \in |R|$. Denote $\mathrm{trdeg}(r/_s k)$ the transcendence degree of r over $\mathrm{Spec}(k)$ via s. Choose an étale morphism $\varphi : V \to R$ where V is a scheme and $v \in V$ mapping to r. Using the definitions mentioned above the lemma we see that

$$\dim_r(R) = \dim_v(V) = \dim(\mathcal{O}_{V,v}) + \mathrm{trdeg}_{s(k)}(\kappa(v)) = \dim(\mathcal{O}_{R,\overline{r}}) + \mathrm{trdeg}(r/_s k)$$

and similarly for t (the second equality by Morphisms, Lemma 02FX). Hence we see that $\mathrm{trdeg}(r/_s k) = \mathrm{trdeg}(r/_t k)$, i.e., (2) holds.

Let k'/k be a field extension. Note that the restriction R' of R to $\mathrm{Spec}(k')$ (see Lemma 04LQ) is obtained from R by two base changes by morphisms of fields. Thus Morphisms of Spaces, Lemma 04NS shows the dimension of R at a point is unchanged by this operation. Hence in order to prove (1) we may assume, by Lemma 04LR, that r is represented by a morphism $r : \mathrm{Spec}(k) \to R$ such that both $s \circ r$ and $t \circ r$ are automorphisms of $\mathrm{Spec}(k)$. In this case there exists an automorphism $R \to R$ which maps r to e (Lemma 04LS). Hence we see that $\dim_r(R) = \dim_e(R)$ for any r. By definition this means that $\dim_r(R) = \dim(R)$.

Part (3) is a formal consequence of the results obtained in the discussion above. □